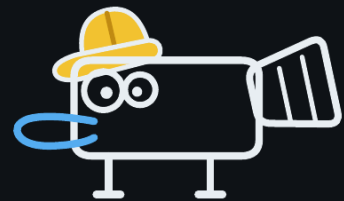


MURDER BOARD · ROUND 1 · FOR PRATEEK

Your questions, answered.

Sir, thank you for the questions. Each section takes one of them in three parts: **what you asked**, **how I answer it**, and **the numbers that justify it**.

- 01 Proto is the moat
- 02 What we actually claim
- 03 Baker first, funded in instalments
- 04 Competition and market
- 05 Automation, honestly
- 06 Competitors' costing
- 07 What it costs, and payback
- 08 Scale
- The vision



proto plans · [Baker](#) makes · [Mixer](#) tests · [Probe Matrix](#) reaches every net
Soham, founder · 2 October 2026 · [protopus-kappa.vercel.app](#)

BEFORE WE START

Sir, none of this is new. It is what we always intended. Your questions help us present it more clearly.

PROTO AT THE CENTRE

Proto was always the heart. [Baker](#), [Mixer](#) and the [Probe Matrix](#) are its hands and eyes in the real world.

BUILT FOR PROTOTYPING

[Baker](#) is for the first 1–10 boards of a design. Volume was always meant for a fab.

THE GAP WE FILL

No comfortable system exists for prototyping, so everyone has to rely on fabs, even for the first board.

BAKER FIRST

₹15 lakh, released in three instalments. Each one is released only after the previous milestone is achieved.

PAYBACK

[Baker](#) pays for itself after 47 boards compared with JLCPCB: about six months for a startup.

HONEST NUMBERS

Prices and performance are projected. The proof of concept measures them.

ABOUT THE NUMBERS

Prices are projected: [Baker](#) ₹2,99,000, [Mixer](#) ₹34,999 (step 2), Proto ₹9,999 a year. The numbers come from PROTOPUS_Economics_Model.xlsx and PROTOPUS_Baker_Mixer_BOM_and_Costing.xlsx (2 Oct 2026). USD is converted at ₹88.

QUESTION 01 · PROTO IS THE MOAT

The moat is Proto. The machines are its hands and eyes.

SIR, YOU ASKED

“The strongest moat is hidden in the idea. If I were the founder, I would pitch Proto instead of two hardware robots, [Baker](#) and [Mixer](#). And we should not project this as a desktop PCB mill, which has a rough history.”

I CAN ANSWER

You’re right, sir. Proto is the company. But an AI can only build hardware if it has hands and eyes in the real world.

Today a commerce graduate can vibe-code an app, because the model reaches every tool it needs through APIs and connectors. Hardware has no API. A model can already draw a board (Flux.ai produces Gerbers), and Claude can drive Fusion 360 and Blender. But no model can etch copper, place a part, or tell whether the I²C bus on a real board is alive. [Baker](#) and [Mixer](#) are that missing API. The [Probe Matrix](#) gives Proto a finger on every net, so it can test and program the board.

And [Baker](#) isn’t a desktop mill. A mill automates one step for people who are already experts. [Baker](#) runs fifteen steps in one machine, with Proto guiding a first-timer through each one.

TO JUSTIFY



AI today: Flux gives Gerbers, Claude drives Fusion and Blender. It stops at the file.

protopus: the AI’s hands and eyes

THE HANDS

[Baker](#)

makes the board: mill, drill, mask, stencil, paste, reflow

THE EYES

[Mixer](#)

signal generator and analysers; replays real sensor faults (step 2)

THE FINGERS

[Probe Matrix](#)

probes every net, so Proto can test and program the board

THE REAL MOAT IS THE DATA

Every run logs design → machine settings → measurement → outcome. A tool that only produces files never sees the real board, so it can never collect this data. The longer Protopus runs, the better Proto gets.

NOT A DESKTOP PCB MILL

A mill automates one step for experts, and that category has a rough history: Bantam Tools stopped selling its PCB mill in 2025. [Baker](#) runs 15 steps in one coordinate frame, from bare FR4 to a reflowed board, and Proto plans each step.

QUESTION 01 · TO JUSTIFY, CONTINUED

How people learn hardware, and why ease is the market

People learn practical skills by doing: try, look, adjust (Kolb’s experiential learning cycle). The speed of that loop sets the speed of learning. Hardware makes a beginner climb eight rungs, and each rung loses people. Proto takes rungs away. Each rung removed brings in a new crowd: students, designers, domain experts and first-time founders.

RUNG	YOU FIRST NEED TO KNOW	WHERE PEOPLE GET STUCK	WHO CARRIES IT NOW
1 · idea → spec	how to state requirements	vague goals, scope creep	Proto asks, then writes the spec
2 · parts	datasheets, footprints, stock	hours of searching; stock runs out	Proto picks in-stock parts
3 · schematic	circuit theory, references	errors that only show on the bench	Proto + Flux check it
4 · layout	an EDA tool, DRC	weeks to learn the tool	Proto fixes it to <u>Baker’s</u> rules
5 · fabrication	Gerbers, vendors, customs	7–20 days to India per try	<u>Baker</u> , on your desk
6 · assembly	SMD soldering, stencils, reflow	skill and equipment	<u>Baker</u> + an auto-generated jig
7 · bring-up	scope, logic analyser, firmware	you can’t tell what’s wrong	bench now; <u>Mixer</u> + <u>Probe Matrix</u> in step 2
8 · iterate	root-cause debugging, DFM	slow loops, fewer attempts	Proto explains the fix

7–20 days

per try today: JLCPCB to India, door to door

On your desk

with Baker: no shipping, customs or minimum order

\$100M ARR

Lovable, 8 months after it removed the coding rung

Rungs 1–4 are already being solved in software. Rungs 5–8 are physical. That is why AI-only tools stop at the file, and why Proto needs machines.

Sources: Kolb (1984) · TechCrunch, 23 Jul 2025 (Lovable) · zbotic.in (JLCPCB to India).

QUESTION 02 · WHAT WE ACTUALLY CLAIM

We never said we replace the fab. We said prototyping has no comfortable home.

SIR, YOU ASKED

“Avoid strong claims like “replaces a fabrication order”. An investor who has invested in hardware manufacturing companies: how would he take it? Softer claims are better, like lower cost, faster lead time, etc.”

I CAN ANSWER

Sir, we never said [Baker](#) replaces a fabrication order. What we said is that no comfortable system exists for prototyping today, so everyone has to rely on fabs, even for the very first board of an idea.

Fabs are built for volume, and they do it well. But a first prototype is a question, not an order: does this circuit work? Today, asking that question means a fab order, 7–20 days of waiting, customs, hand assembly and a borrowed bench.

Think of a plastic bottle. You 3D-print it to check its shape, feel and fit, and only then pay for the injection mould. Plastic has had that comfortable desktop path for years. Electronics never got one that ends in a finished, soldered, tested board. [Baker](#) is that path. Once a design is frozen, Proto hands the fab files that have already worked on a real board. **So an investor in manufacturing gets a company that sends the fabs better first orders.**

TO JUSTIFY

WHAT WE SAY	WHAT IT MEANS	BASIS
No comfortable system exists for prototyping	a first board today means a fab order, hand assembly and a borrowed bench	₹6,721 per assembled try from JLCPCB to India, 7–20 days
Lower cost per try	about ₹334 of materials per board on Baker	Board Cost sheet, Economics Model
Shorter lead time	the board is made on your desk: no shipping, customs or minimum order	the proof of concept measures it
Built for prototypes	the first 1–10 boards of a design; volume goes to a fab	see question 8

THE PATH FROM IDEA TO 1,000S

	PROTOTYPE	VOLUME
a plastic bottle	desktop 3D print: shape, feel, fit	injection mould
a circuit board, today	fab order + hand assembly + a borrowed bench	fab / EMS
a circuit board, with us	Proto + Baker : a finished board on the desk	fab / EMS, with files already proven on a real board

QUESTION 03 · ONE PRODUCT AT A TIME

Baker first, funded in instalments.

SIR, YOU ASKED

“Two products for a first-shot raise of ₹10L is, in my opinion, a bit aggressive. We should go step by step. Maybe start with Baker first?”

I CAN ANSWER

Agreed, sir. We will go one product at a time, starting with Baker and Proto.

The two machines are the vision: one makes the hardware, the other tests it, programs it and feeds the results back to Proto. But the money we raise now builds only Baker. While we prove it, its boards are tested on ordinary bench instruments. Mixer and the Probe Matrix come in step 2, funded by the seed round after the proof of concept.

To keep it step by step for you too, I suggest the funding comes in **instalments**. Each instalment is released only after the previous milestone is achieved and shown to you.

TO JUSTIFY

STEP 1 · NOW · ₹15L

Proto + Baker

all 15 steps running, 20 proof-of-concept boards, Proto from a brief to a Baker job

STEP 2 · SEED · ~₹1.5 CR

Mixer + Probe Matrix

test and program every board; pilot units built turnkey in China

STEP 3 · ECOSYSTEM

Proto runs it all

Baker, Mixer, the Probe Matrix and any 3D printer; India first, then global

WHERE ₹15,00,000 GOES · 9 MONTHS · NO FOUNDER SALARY

LINE	₹
<u>Baker</u> build, costed line by line (70-line BOM, duty, 15% contingency)	3,73,418
<u>Baker</u> Rev B + ~150 test boards of materials	1,60,000
one software engineer who also writes firmware, 9 × ₹50,000	4,50,000
Proto software + compute, 9 × ₹15,000	1,35,000
testing, tooling, lab safety	1,50,000
IP + legal, outreach, reserve	2,31,582
Total	15,00,000

QUESTION 03 · SUGGESTED: FUNDING IN INSTALMENTS

The next instalment is released only when the previous milestone is achieved.

Sir, rather than releasing ₹15 lakh at once, I suggest three instalments over nine months. You see each milestone before the next money moves, so the risk you take is always tied to progress you can see. The amounts follow the same budget, line by line.

INSTALMENT 1

₹6,00,000

at signing · months 1-2

- [Baker](#) build: ₹3,73,418
- engineer, 2 months: ₹1,00,000
- Proto compute, 2 months: ₹30,000
- tooling and lab set-up: ₹75,000
- IP filing: ₹21,582

MILESTONE 1 · MONTH 2

[Baker](#) runs all 15 steps on one board, shown to you on video and in person.

↓ next instalment released only after this milestone is shown

INSTALMENT 2

₹5,00,000

after milestone 1 · months 3-5

- [Baker](#) Rev B + ~150 test boards: ₹1,60,000
- engineer, 3 months: ₹1,50,000
- Proto compute, 3 months: ₹45,000
- testing: ₹50,000
- IP + legal: ₹50,000 · outreach: ₹45,000

MILESTONE 2 · MONTH 5

Process tests pass (mask, stencil, jig accuracy). Proto goes from a plain-English brief to Gerbers to a [Baker](#) job. [Baker](#) Rev B is built.

↓ next instalment released only after this milestone is shown

INSTALMENT 3

₹4,00,000

after milestone 2 · months 6-9

- engineer, 4 months: ₹2,00,000
- Proto compute, 4 months: ₹60,000
- testing: ₹25,000
- IP, outreach and reserve: ₹1,15,000

MILESTONE 3 · MONTH 6

Proof of concept: 20 boards made with Proto + [Baker](#) and measured. Then the seed round, months 7-9.

IF THE ROUND IS ₹10 LAKH

If the round is capped at ₹10 lakh, the same three milestones apply, with instalments of roughly ₹5L, ₹3L and ₹2L. That version covers [Baker](#), Rev B, six months of the engineer, compute and testing, with no lab set-up, IP or outreach.

QUESTION 04 · COMPETITION AND MARKET

Everyone else solves one step. We run the whole loop.

SIR, YOU ASKED

“I think a thorough competition analysis and market study is needed.”

I CAN ANSWER

We did the study, sir, and here it is in one paragraph.

Protopus sits where two large markets meet: printed circuit boards (₹6.4 lakh crore worldwide; ₹55,000 crore in India, growing about 15% a year) and test and measurement (₹3.1 lakh crore worldwide; ₹8,300 crore in India). Our first market is much smaller, and we can count it. India has 5,868 AICTE engineering colleges and about 2,250 hardware startups: 8,118 buyers, or about ₹243 crore of Baker sales at one machine each, plus ₹8.1 crore a year in Proto plans. Behind them come 10,000 Atal Tinkering Labs, with 50,000 more announced. **Every competitor solves one step.** A mill makes a bare board for an expert. A placer only places parts. A USB instrument measures, but a human has to read the screen. A fab ships in weeks, and an AI design tool stops at the file. **Nobody runs the whole loop, and nobody lets an AI drive the machines. That is where we are different.**

TO JUSTIFY

₹6.4L cr	₹3.1L cr	₹55K cr	₹8.3K cr
global PCB market	global test + measurement	India PCB, ~15% a year	India test + measurement
INDIA BEACHHEAD	BUYERS	BAKER (₹2,99,000)	WITH MIXER (₹3,33,999)
AICTE engineering colleges	5,868	₹175 Cr	₹196 Cr
hardware startups (1% of 2.25 lakh DPIIT)	2,250	₹67 Cr	₹75 Cr
Total, one machine each	8,118	₹243 Cr	₹271 Cr

Plus ₹8.1 Cr a year of Proto plans (₹9,999 × 8,118). Next: 10,000 Atal Tinkering Labs (₹20 lakh each over five years; 50,000 more planned), ~2,500 Fab Labs worldwide, 33M Arduino users and 2.3M Hackster members.

	MAKE	ASSEMBLE	TEST	AI GUIDE
Desktop mills (Makera Carvera, LPKF)	●	○	○	○
Ink printer (Voltera V-One)	●	●	○	○
Pick-and-place (Neoden YY1)	○	●	○	○
USB instruments (Digilent AD3, Moku:Go)	○	○	●	○
Fabs (JLCPCB, PCBWay)	●	●	○	○
AI design (Flux.ai)	○	○	○	●
protopus: Proto + Baker (+ Mixer)	●	●	●	●

Sources: IMARC · Grand View · AICTE · DPIIT/TiE (2.25 lakh startups; the 1% hardware share is our assumption) · Atal Innovation Mission · Union Budget 2025-26 · Fab Foundation · Arduino · Hackster.

QUESTION 05 · AUTOMATION, HONESTLY

Not 100% automatic, but you don't load a board every time.

SIR, YOU ASKED

“Send the design at night, pick it up in the morning’ is a bit too aggressive a claim. We have to load boards, place components, etc. So it’s not fully automated.”

I CAN ANSWER

You’re right, sir, it isn’t 100% automatic.

But you don’t have to load a board each time. A refill magazine feeds the next blank, and you only top it up now and then.

Reflow and jig-making happen inside the machine, but they are not fully automated. The placement jig is generated automatically from the Gerber and printed. You fill it with the components and place it for reflow, and the reflow profile then runs by itself.

TO JUSTIFY

STEP	WHAT HAPPENS	WHO DOES IT
01-02	feed the blank from the refill magazine, probe the height map	machine
03	isolation mill + drill	machine
04-09	clean, coat the mask, UV image, develop, cure	machine
10-12	laser-cut the stencil, clamp, paste, release	machine
jig	generated from the Gerber and printed inside Baker , in parallel with steps 01-12	machine
13	you fill the jig with the components	you
14	you place the jig for reflow; the 138 °C profile runs itself	you + machine
15	test: Proto runs the plan (bench instruments now; Probe Matrix in step 2)	you connect it

Steps 1–12

run on their own, from blank to pasted board

Refill magazine

no loading a board every time

Auto jig

generated from the Gerber; you fill it

QUESTION 06 · COMPETITORS' COSTING

The same loop costs ₹7.8 lakh elsewhere, and still has no AI.

SIR, YOU ASKED

“I also strongly recommend, as mentioned above, looking at the competition and their costing.”

I CAN ANSWER

Here is what the same loop costs elsewhere, sir.

Bought as separate machines (a mill, a placer and a USB instrument), it comes to about ₹7.82 lakh before import duty, and it still has no solder mask, no stencil and no AI. [Baker](#) with Proto is ₹2.99 lakh; with [Mixer](#), ₹3.34 lakh. Our production cost is ₹1,44,842 for [Baker](#) and ₹13,740 for [Mixer](#).

TO JUSTIFY

PRODUCT	WHAT IT DOES	LIST PRICE	WHERE IT LOSES TO US
LPKF ProtoMat	professional PCB mill	₹8.6–24.5L per machine	separate machines for mask, plating, assembly; for experts
Voltera V-One	prints conductive ink; dispenses paste	~₹3.1L (\$3,500)	ink traces, not etched copper; no mask, stencil or test
Makera Carvera / Air	desktop CNC mill	₹4.8L / ₹2.2L	mills only
Bantam Tools PCB mill	desktop PCB mill	discontinued 2025	the “rough history”: one step, for experts
Neoden YY1	pick-and-place	~₹2.6L	places parts only
Digilent AD3	USB scope + logic + supplies	~₹34,000	a human reads the screen; it doesn't know the design
JLCPCB / PCBWay	fab + assembly	₹6,721 per try, landed	7–20 days; minimum orders; customs
Flux.ai	AI PCB design (SaaS)	₹1,300–4,300 / editor / month	stops at the file; Proto uses its API and goes on
protopus: Proto + Baker	board, mask, stencil, paste, reflow; AI-planned	₹2,99,000 + ₹9,999/yr	Mixer ₹34,999 + Probe Matrix in step 2

THE SAME LOOP, BOUGHT SEPARATELY, VS PROTOPUS

mill + placer + AD3		₹7.82L: no mask, stencil or AI
Baker + Mixer		₹3.34L: the whole loop
Proto + Baker		₹2.99L: step 1

List prices at ₹88 per US\$, before Indian duty and GST. Sources: LDKF via Design World · Voltera · Makera · ToolGuyd · EEVblog · RS/RobotShop · Liquid Instruments · Red Pitaya · Mouser · JLCPCB + zbotic.in · flux.ai. Protopus prices projected.

QUESTION 07 · WHAT IT COSTS, AND PAYBACK

₹2,99,000. It pays for itself after 47 boards.

SIR, YOU ASKED

“How much is this machine going to cost? Say X lacs: we also need to see how many boards are created and whether it cancels my cost.”

I CAN ANSWER

Baker will be ₹2,99,000, sir (projected, before GST).

On Baker, a board costs about ₹334 in materials: copper-clad board, mask, paste, stencil film, jig resin, bit wear and electricity. The same assembled board from JLCPCB, landed in India with shipping and customs, costs ₹6,721 and takes 7–20 days to arrive. So Baker cancels its cost after **47 boards**. That is about six months for a startup making two boards a week, and about a year for a college lab making one.

This count is conservative. It leaves out the engineer's time and the days spent waiting, which are usually worth more than the boards.

TO JUSTIFY

₹2,99,000		₹334	₹6,721	47 boards	
<u>Baker</u> list price (projected)		materials per board on <u>Baker</u>	same board, JLCPCB landed in India	to pay <u>Baker</u> back	
CUSTOMER BUYS	COMPARED WITH	SAVING / BOARD	BOARDS TO PAY BACK	STARTUP, 2 A WEEK	LAB, 1 A WEEK
<u>Baker</u>	JLCPCB to India	₹6,387	47	5.9 months	11.8 months
<u>Baker</u>	Indian 24-h fab	₹5,166	58	7.2 months	14.5 months
<u>Baker</u> + <u>Mixer</u>	JLCPCB to India	₹6,387	53	6.6 months	13.2 months
IF THE BAKER PRICE MOVES		₹2,99,000	₹3,49,000	₹3,99,000	₹4,49,000
boards to pay back <u>Baker</u>		47	55	63	71
boards to pay back <u>Baker</u> + <u>Mixer</u>		53	61	68	76
our gross margin on <u>Baker</u>		51.6%	58.5%	63.7%	67.7%

FROM OUR SIDE

Each Baker earns ₹1,54,158 of gross profit (51.6%), or about ₹99,000 after all costs. So 10 Bakers sold return the whole ₹15 lakh round on gross profit, and 16 do so on all-in profit.

JLCPCB: 5 PCBs + PCBA (\$8 setup, \$1.50 stencil) + DHL + 10% BCD + 18% IGST. Adding the first year of the Proto plan (₹9,999) moves Baker's payback to 49 boards.

QUESTION 08 · SCALE

Baker scales the number of people who can prototype, not boards per machine.

SIR, YOU ASKED

“Innovation is good but may not be scalable in a production environment. 1,000 boards? 10,000? 1 lakh?”

I CAN ANSWER

Baker isn't meant for production, sir, and we won't claim it is. It is for the first one to ten boards of a design.

At 10,000 or a lakh boards, a fab is the right tool, and Proto hands it files that have already worked on a real board. What scales is the number of people who can prototype. Proto is software, so each extra user costs only compute. Our own manufacturing scales too: it is turnkey in China, with final test, minimal assembly and shipping from India.

TO JUSTIFY

BOARDS A YEAR	RIGHT TOOL	WHY
10	<u>Baker</u>	prototypes on your desk beat fab minimums and shipping
100	<u>Baker</u>	one machine serves a busy lab or startup
1,000	<u>Baker</u> + a fab	iterations on <u>Baker</u> ; any batch of 10+ identical boards goes to a fab
10,000	fab / EMS	production; Proto exports the checked Gerber, BOM and placement files
1,00,000	fab / EMS	not what <u>Baker</u> is for, and we don't claim it

HOW WE BUILD

China, turnkey

parts sourced and assembled in China, then factory-tested

HOW WE SHIP

India

final test, minimal assembly and shipping; India first, then global

HOW PROTO SCALES

Software

each extra user costs only compute; every board makes it better

THE NUMBERS BEHIND IT

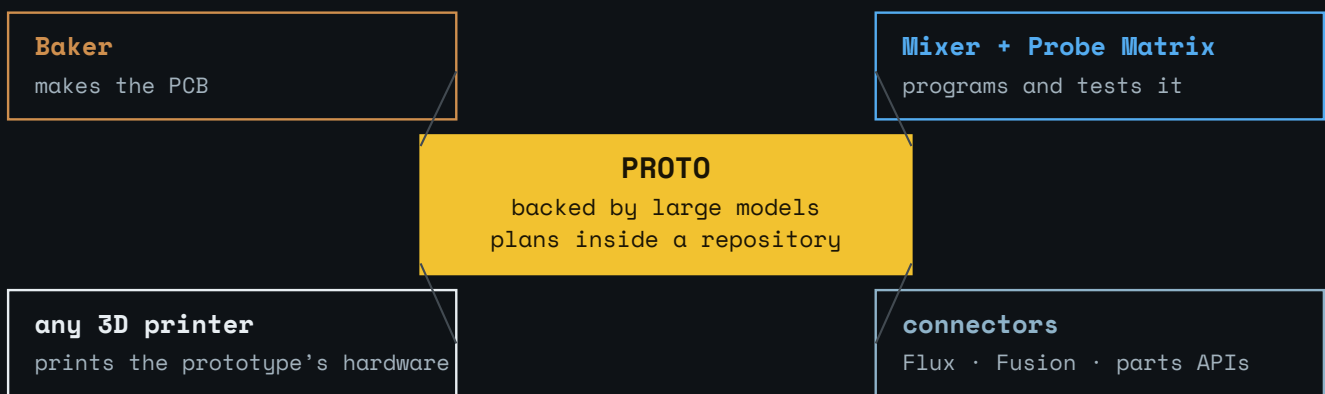
Production cost is ₹1,44,842 per Baker at 100 units, with RFQs planned from 5 to 5,000 units. The China master plan covers the plant, the suppliers and the quality gates.

THE VISION

Sir, here is where this is going: from idea to prototype, without the friction.

Today there is good compute and really good reasoning models, and they are very good at connecting different apps to get a task done. That is why a commerce graduate can vibe-code a software product. The AI part is proven: Flux.ai gives Gerbers, and Claude connectors drive Fusion 360 and Blender. **But for a hardware product, the reasoning model needs a way into the real world: smart machines and smart accessories.** Nobody has put the two together yet.

Now imagine an ecosystem where Proto plans everything from a single input and, step by step, takes over the machines, including a 3D printer. It maps out the hardware and the software behind it, tests it with real signals, flags what is wrong and guides you to the fix. The learning curve becomes so easy that going from idea to innovation feels effortless.



We will never sell a 3D printer: Proto drives the one you already have.

THE VISION · HOW IT WORKS

One input from you. Proto plans the rest.

PROTO, BACKED BY LARGE MODELS

Large reasoning models break your goal into steps, call the right tool for each and check the result. Every plan lives in a project repository: spec, schematic, Gerbers, firmware, printed parts and test results, versioned like code.

ELECTRONICS

Proto designs the PCB, picks the components and sensors, and produces the BOM with affiliate purchase links for every part.

MECHANICAL

Proto designs the prototype's hardware for 3D printing, sized from measurements of the real board, taken in the same environment.

TEST AND FEEDBACK

[Mixer](#) with the [Probe Matrix](#) programs the board, tests it with real signals, flags what failed and gives a plan of action.

THE VISION · HOW PROTO 3D-PRINTS THE HARDWARE

From the measured board to printed parts that fit.

The PCB already tells Proto most of what the hardware needs. The outline and mounting holes come from the Gerber. Every component's position comes from the placement file. Their heights come from the parts' 3D models, checked against the jig [Baker](#) printed. The buttons, connectors and sensors fixed on the board set where the openings go. Proto turns that into printable parts in five steps:

01 · MEASURE

Proto collects the board outline, holes, component positions and heights, and the positions of buttons, ports and sensors. Everything is measured on the real board, in the same environment.

02 · MODEL

Proto writes the parts as parametric CAD through the Fusion 360 or Blender connector, or as code-based CAD (OpenSCAD / CadQuery). This gives an enclosure, mounts, brackets and button caps, with clearances and snap-fits built in. You can change any dimension in plain English.

03 · SLICE

Proto exports STL or 3MF and runs a slicer (PrusaSlicer, OrcaSlicer or Cura) with your printer's profile and material. It chooses orientation, supports and infill for strength where the parts need it.

04 · PRINT

Proto sends the G-code to your printer over its own network interface (OctoPrint, Klipper/Moonraker, PrusaLink or Bambu LAN mode), or gives you the file to print by hand. It never needs to be our printer.

05 · FIT AND IMPROVE

You assemble the board into the printed parts and tell Proto what felt off: too tight, a stiff button, a gap. Proto changes the parameters, and the next print fixes it. Every version stays in the project repository.

IN THE SMARTWATCH EXAMPLE

For the smartwatch, Proto would print the case, the strap mounts and ten button caps. The caps line up with the ten switches on the PCB, and the case has a window for the heart-rate sensor. The board outline and the switch positions come straight from the Gerber and the placement file, so the first print is already close.

The CAD connectors and printer interfaces named here already exist. Joining them into one flow with [Baker](#) and [Mixer](#) is the work ahead, planned for step 3.

THE VISION · AN EXAMPLE

A smartwatch with 10 physical buttons, from one sentence.

- 01 You describe the smartwatch in plain English: ten physical buttons, the sensors you want, how long it should last.
- 02 Proto checks feasibility and limits: what is possible at this size, and what isn't.
- 03 Proto proposes the best prototype model for your goal.
- 04 Proto writes the requirements back in plain English for you to approve.
- 05 Proto designs the PCB, picks the components and sensors, and builds the BOM with purchase links.
- 06 [Baker](#) makes the PCB. Your 3D printer prints the watch's hardware: the case, the strap mounts and the ten button caps, sized from the measured board.
- 07 Proto programs the board through [Mixer](#) and the [Probe Matrix](#).
- 08 [Mixer](#) and the [Probe Matrix](#) test every button and sensor with real signals, flag any failure and give you a plan of action.
- 09 You wear it and tell Proto what felt off: a button too stiff, a screen too dim.
- 10 Proto improves the next iteration, and the loop runs again until you are ready for a fab.

The one-stop solution for prototyping.

WHO WE ARE

Protopus is a company that innovates smart solutions for prototyping, with smart machines and accessories. [Baker](#) and [Mixer](#) are our first products. Our software and hardware teams keep improving both. Manufacturing is fully turnkey in China; we test it here with minimal assembly, and ship first in India, then globally.

Thank you, sir.

Happy to get on a call and go through any of this in detail.

[website](#) · [Baker](#) · [Mixer](#) · [Probe Matrix](#)

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